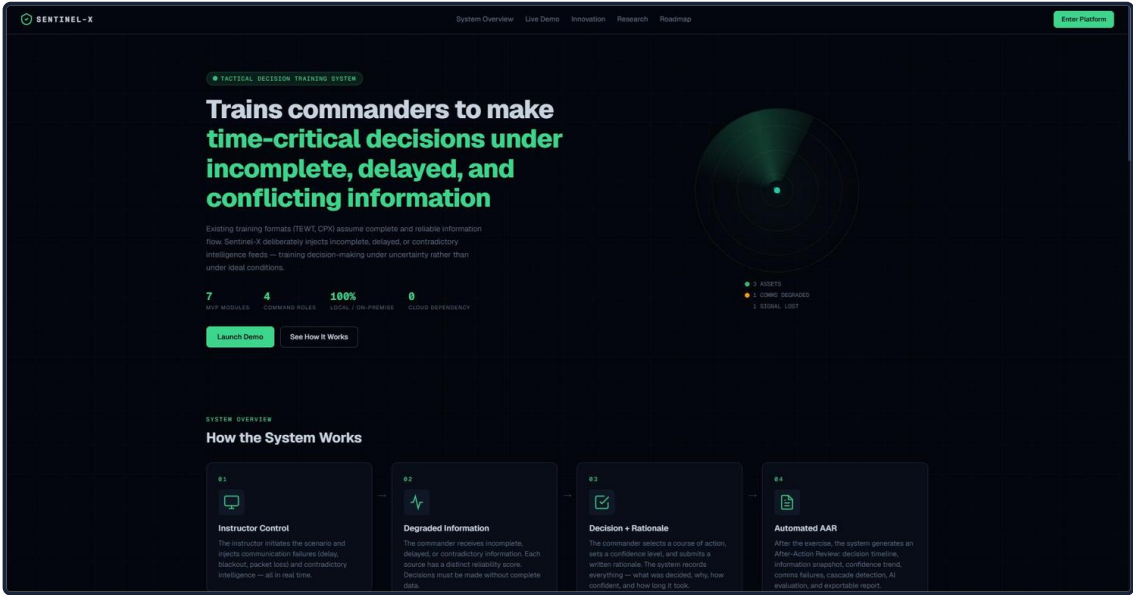


SENTINEL-X

Multi-Domain Decision Trainer for Degraded Communication Environments

USER MANUAL

Version 5.0 | Fullstack Edition



Landing Page — Home

Landing Page — Home

Top Navigation Bar

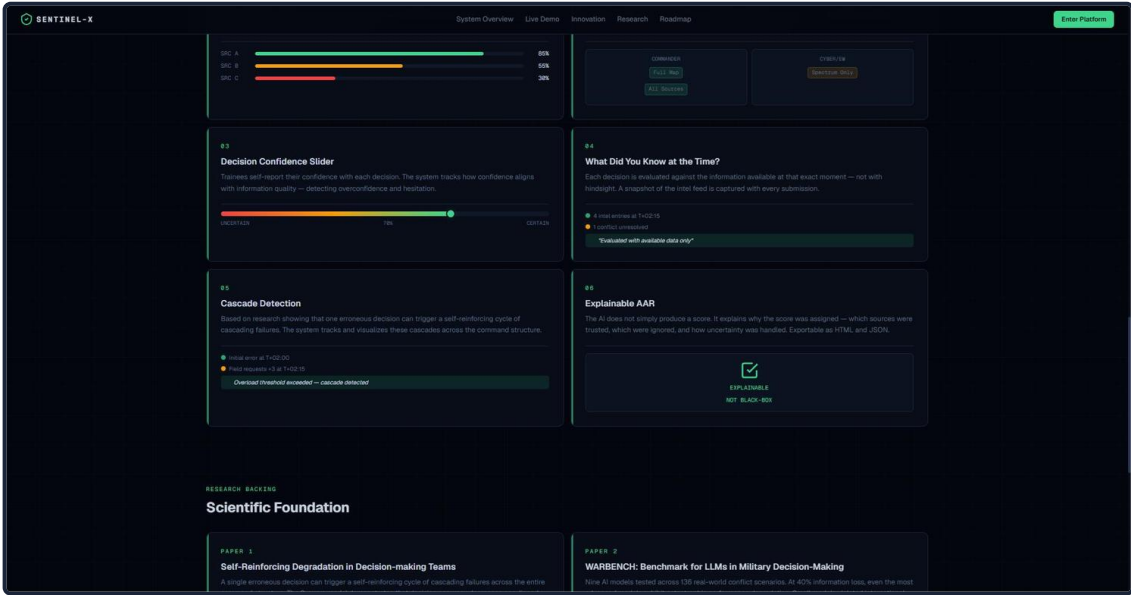
Baayein taraf SENTINEL-X logo. Navigation links: System Overview, Live Demo, Innovation, Research, Roadmap. Right mein 'Enter Platform' button — is par click karke login/register page par jaayein.

Hero Section

Main headline: 'Commanders ko incomplete, delayed, aur conflicting information ke under time-critical decisions lene ki training deta hai'. Key stats: 7 MVP Modules, 4 Command Roles, 100% Local/On-Premise, 0 Cloud Dependency.

Tactical Radar Visual

Right side mein ek dark circular tactical radar display hai with active asset status indicators (3 Assets, 1 Comms Degraded, 1 Signal Lost).



Landing Page — Innovation Section

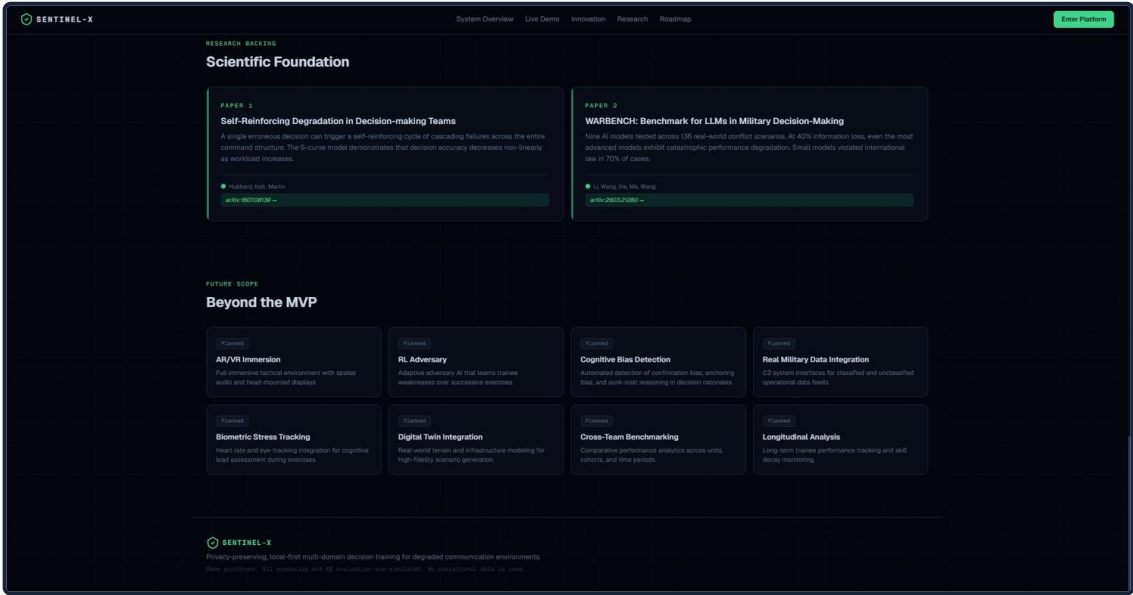
Landing Page — Innovation Section

System Kaise Kaam Karta Hai

Char sequential process cards: 01 Instructor Control (real-time communication failures inject karta hai), 02 Degraded Information (commander ko incomplete/conflicting data milta hai), 03 Decision + Rationale (course of action selection, confidence levels), 04 Automated AAR (post-exercise After-Action Review generate karta hai).

Live Demo Section

Interactive demo tabs: Normal Comms, Comms Degraded, Conflicting Intel, Decision, aur AAR phases. Niche 'Launch Full Simulation' button hai.



Landing Page — Research & Roadmap

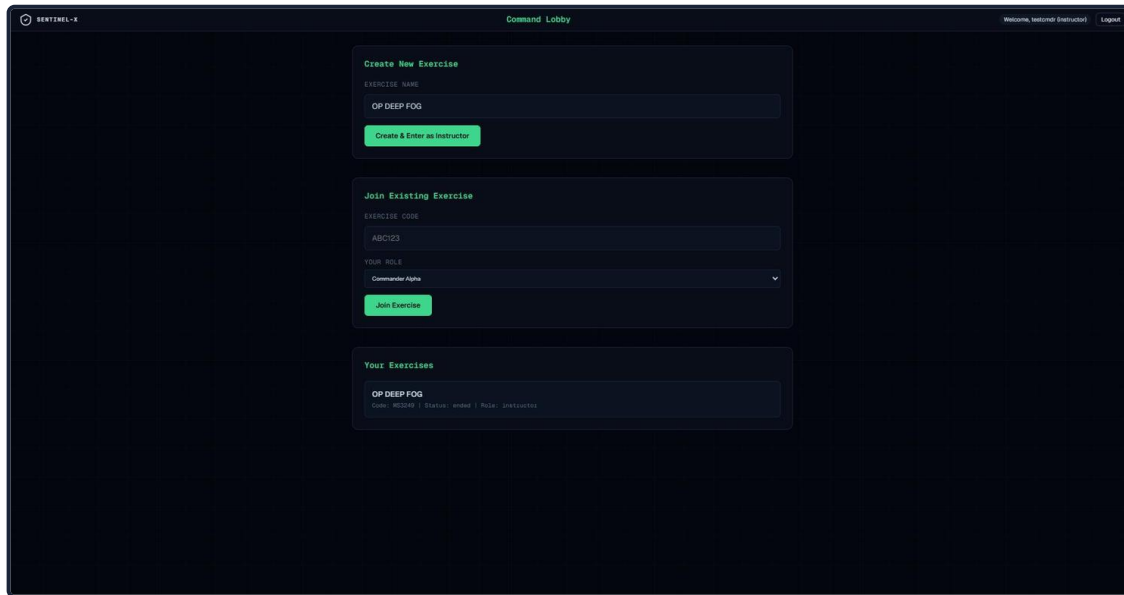
Landing Page — Research & Roadmap

Research Backing

Academic papers jo system design ko inform karte hain. arXiv:1607.08139 (Hubbard, Kott, Martin dwara decision-making teams mein self-reinforcing degradation) aur arXiv:2603.21280 ke links. Ye papers cascade detection aur degraded communication training ke theoretical basis hain.

Roadmap Section

Development roadmap dikhata hai with future milestones aur Sentinel-X platform ke liye planned features.



Command Lobby — Exercise Create & Join

Command Lobby — Exercise Create & Join

Header

'Welcome, testcmdr (Instructor)' dikhta hai Logout button ke saath. Command Lobby title green colour mein displayed hai.

Create New Exercise

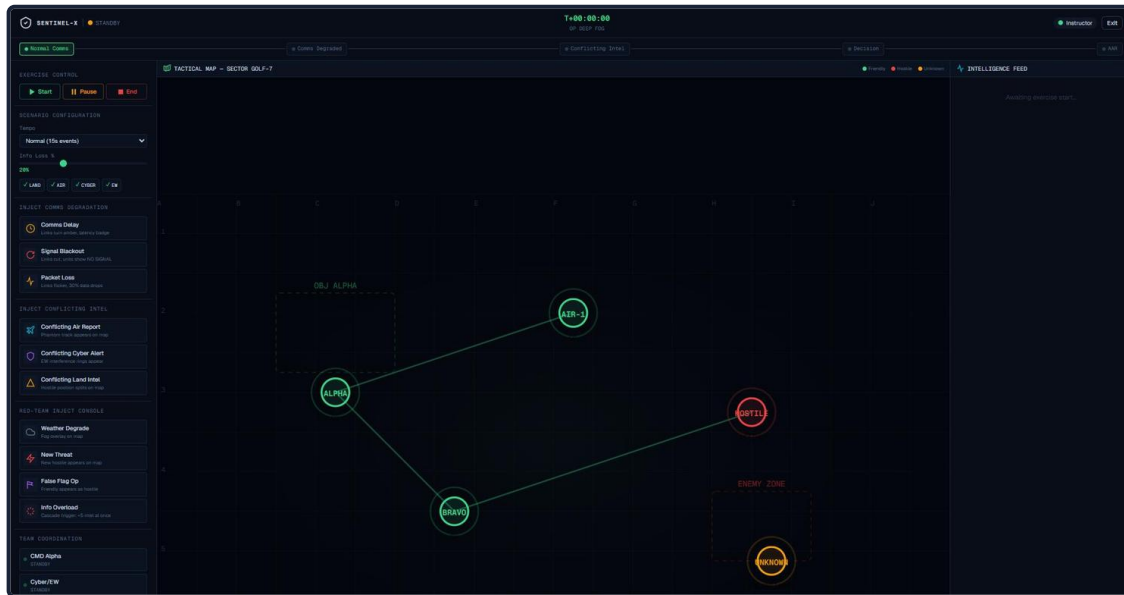
EXERCISE NAME field mein exercise naam likhein (jaise OP DEEP FOG) aur 'Create & Enter as Instructor' par click karein. System ek unique code generate karta hai aur aap instructor banke simulator mein enter karte hain.

Join Existing Exercise

Apne instructor dwara diye gaye exercise code ko likhein, dropdown se apna role select karein (Commander Alpha, Cyber/EW Officer, Air Liaison, ya Intel Analyst) aur 'Join Exercise' par click karein.

Your Exercises

Aapke dwara create ya join kiye gaye saare exercises ki list — naam, code, status (created/active/ended), aur aapka role ke saath.



Simulator — Standby Mode (Start se pehle)

Simulator — Standby Mode (Start se pehle)

Top Status Bar

SENTINEL-X logo with STANDBY status badge. Exercise timer T+00:00:00 dikhata hai. Exercise naam (OP DEEP FOG) displayed. Role indicator 'Instructor' dikhata hai. Right mein Exit button.

Phase Tabs

Paanch tabs exercise progression dikhate hain: Normal Comms (active), Comms Degraded, Conflicting Intel, Decision, aur AAR.

Left Sidebar — Exercise Control

Start, Pause, aur End buttons. Scenario Configuration: Tempo dropdown (Slow/Normal/Fast), Info Loss % slider, Domain checkboxes (Land, Air, Cyber, EW).

Left Sidebar — Inject Controls

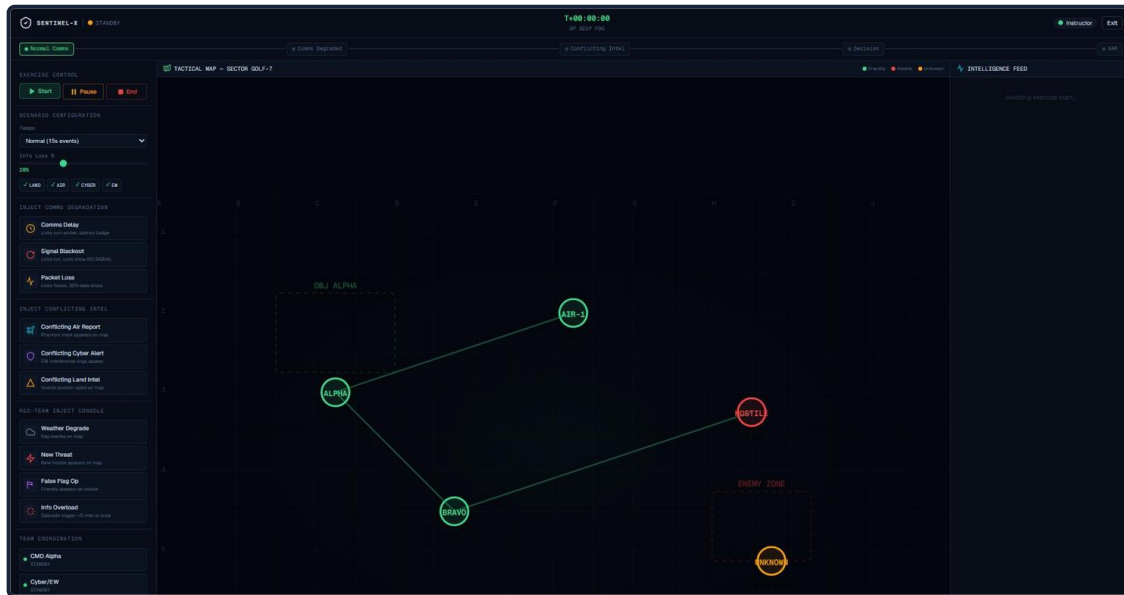
Teen groups: Inject Comms Degradation (Comms Delay, Signal Blackout, Packet Loss), Inject Conflicting Intel (Air Report, Cyber Alert, Land Intel), Red-Team Inject Console (Weather Degrade, New Threat, False Flag Op, Info Overload).

Center — Tactical Map

Sector Golf-7 ka grid map coordinate labels A-J aur 1-5 ke saath. Friendly units (ALPHA, BRAVO, AIR-1 green), hostile units (HOSTILE red), unknown units (UNKNOWN, UNCONFIRMED yellow), objective zones (OBJ ALPHA, ENEMY ZONE) dikhata hai.

Right — Intelligence Feed

Exercise start hone tak 'Awaiting exercise start...' dikhata hai.



Simulator — Exercise Active (Normal Comms)

Simulator — Exercise Active (Normal Comms)

Status Change

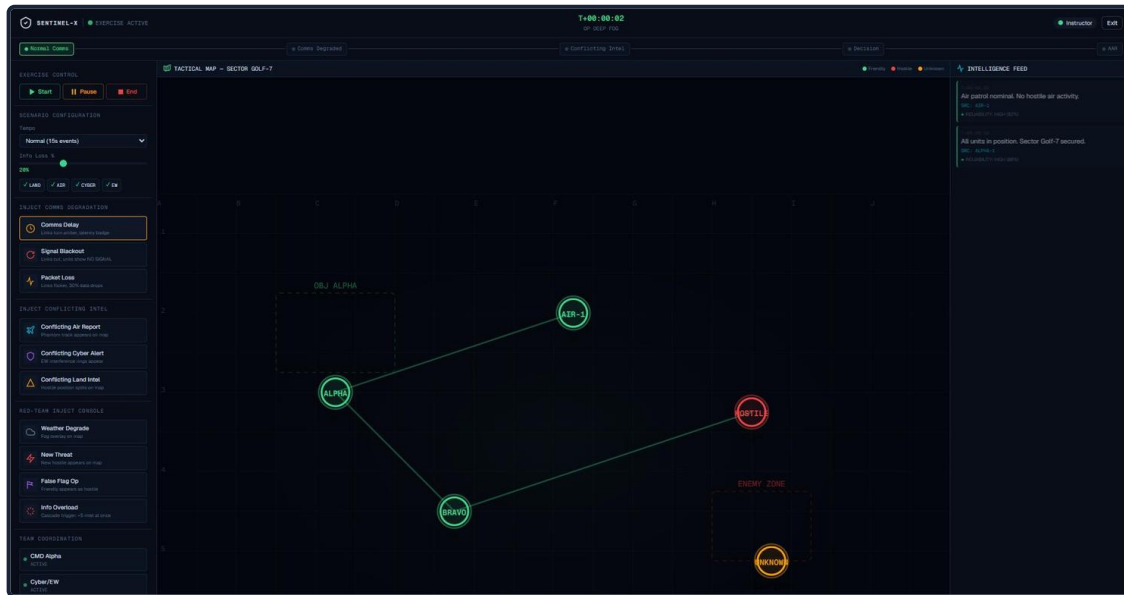
Status badge STANDBY se EXERCISE ACTIVE (green) mein change hota hai. Timer count karna shuru karta hai. Team coordination roles (CMD Alpha, Cyber/EW) ACTIVE ho jate hain.

Intel Feed Populated

Intelligence Feed baseline reports receive karna shuru karta hai: 'Air patrol nominal', 'All units in position'. Har entry mein timestamp, message, source, aur reliability score (HIGH/MEDIUM/LOW) hota hai.

Map State

Tactical map units ke beech normal green communication links dikhata hai. Saare friendly units (ALPHA, BRAVO, AIR-1) connected aur operational hain.



Simulator — Comms Degradation Injected

Simulator — Comms Degradation Injected

Comms Delay Effect

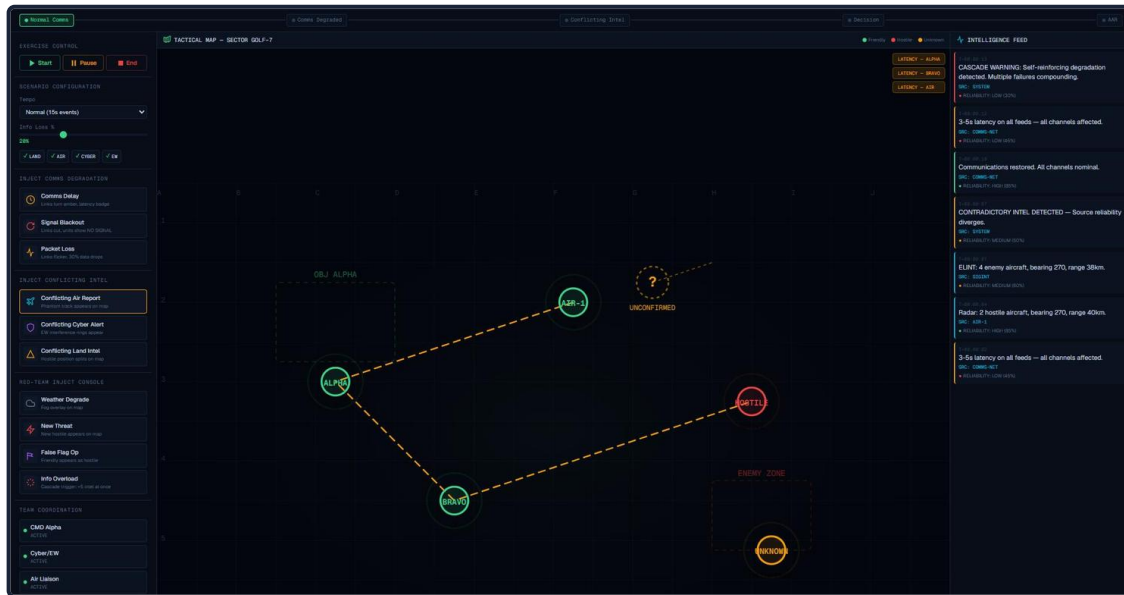
Instructor ne 'Comms Delay' par click kiya. Tactical map par network links dashed amber lines ban jaate hain. Yellow latency badges (LATENCY - ALPHA, LATENCY - BRAVO, LATENCY - AIR) map par appear hote hain.

Intel Feed Warning

Intelligence Feed log karta hai: '3-5s latency on all feeds — all channels affected'. Ye real-world communication delays ko simulate karta hai degraded environments mein.

Auto-Recovery

Lagbhag 8 second baad degradation automatically recover hota hai. Ek 'COMMS RESTORED' message intel feed mein inject hota hai aur links wapas green ho jaate hain.



Simulator — Conflicting Intel & Cascade Warning

Simulator — Conflicting Intel & Cascade Warning

Phantom Contact

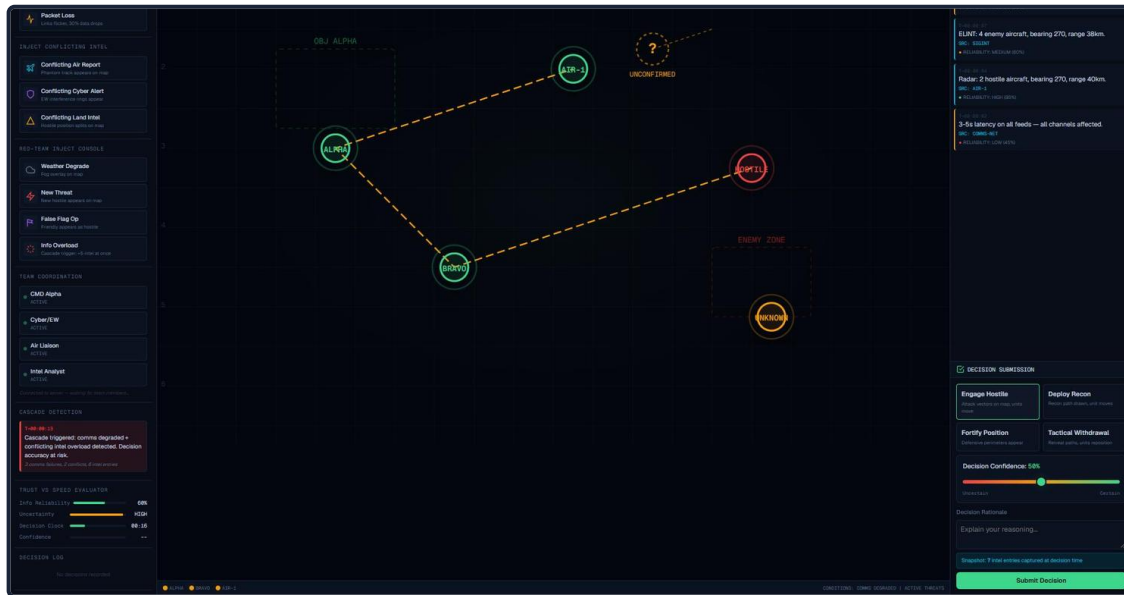
'Conflicting Air Report' inject karne ke baad, tactical map par AIR-1 ke paas ek dashed unconfirmed phantom contact '? UNCONFIRMED' appear hota hai. Ye divergent sensor data represent karta hai.

Diverging Intel

Intelligence Feed contradictory reports dikhata hai: Radar 2 hostile aircraft report karta hai, jabki ELINT 4 enemy aircraft report karta hai. Reliability scores alag hain, commander ko assess karna padta hai kaunse source par bharosa karein.

Cascade Warning

Jab 2+ comms channels degraded hain aur 1+ conflicting intel present hai, system automatically cascade condition detect karta hai aur intel feed mein 'CASCADE WARNING: Self-reinforcing degradation detected' inject karta hai.



Decision Submission & Confidence Slider

Decision Submission & Confidence Slider

Decision Options

Chaar course-of-action buttons: Engage Hostile (attack vectors draw hote hain, units engage karne move karte hain), Deploy Recon (recon path draw hota hai, unit investigate karne move karta hai), Fortify Position (defensive perimeters appear hote hain), Tactical Withdrawal (retreat paths draw hote hain, units reposition karte hain)।

Confidence Slider

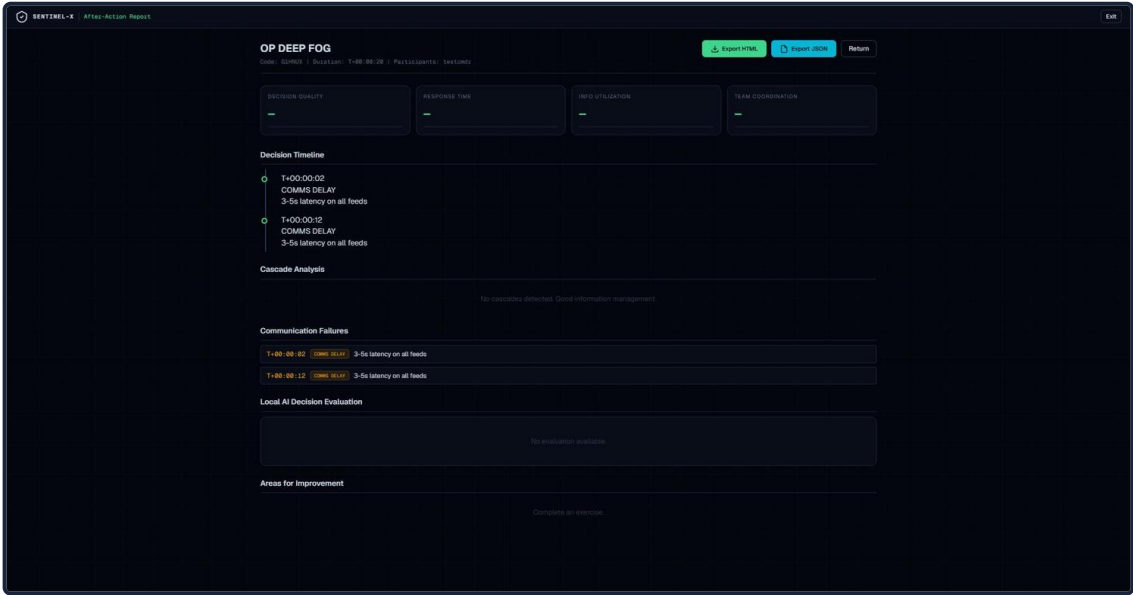
Decision select karne ke baad ek confidence slider appear hota hai (0-100%)। Trainee apna confidence level set karta hai. System overconfidence (limited intel par high confidence) aur hesitation (ample intel par low confidence) track karta hai।

Rationale Input

Trainee ke decision ke liye written rationale ka ek text area। Ye submission ke moment par ek intel snapshot ke saath capture hota hai — 'What Did You Know at the Time?' evaluation ko support karta hai।

Cascade Detection Panel

Real-time cascade status dikhata hai। Trust vs Speed Evaluator Info Reliability %, Uncertainty level, aur Decision Clock display karta hai।



After-Action Review (AAR) — Timeline & Analysis

After-Action Review (AAR) — Timeline & Analysis

AAR Header

Exercise naam (OP DEEP FOG), code, duration (T+00:00:20), aur participants dikhata hai. Top right mein Export HTML, Export JSON, aur Return buttons.

Score Cards

Chaar metric cards: Decision Quality, Response Time, Info Utilization, Team Coordination — har ek exercise performance par based 0-100 score hota hai.

Decision Timeline

Saare events ka chronological view: comms delays, conflicting intel, decisions made, aur cascades detected. Har event mein timestamp aur description hota hai.

Cascade Analysis

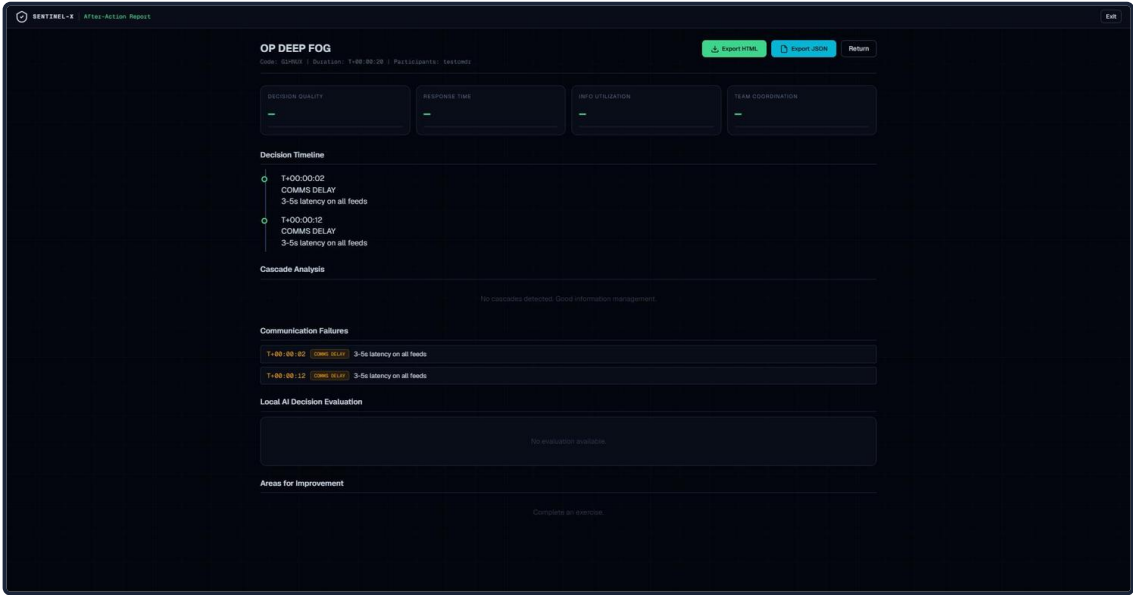
Detected cascades ki detail trigger descriptions ke saath. Agar information well-managed rahi toh 'No cascades detected' dikhata hai.

Communication Failures Log

Saare injected degradation events timestamps ke saath — comms delays, blackouts, packet losses.

AI Evaluation & Improvement

Local AI evaluation overall rating (EXCELLENT/PROFICIENT/DEVELOPING/NEEDS RETRAINING) findings aur personalized recommendations ke saath deta hai.



AAR — Export Options

AAR — Export Options

Export HTML

Tables, timelines aur score cards ke saath ek fully formatted report download karne ke liye 'Export HTML' par click karein. Ye ek printable, shareable document banata hai.

Export JSON

Raw structured data download karne ke liye 'Export JSON' par click karein. Dusre systems ke saath integration, programmatic analysis, ya archival ke liye useful.

Return Button

Command Lobby par wapas jaane ke liye 'Return' par click karein jahan aap naya exercise create kar sakte hain ya kisi aur mein join kar sakte hain.